

The Language of Medicine

13th edition

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Chapter 5

Digestive System

Chapter Goals

- Name the organs of the digestive system and describe their locations and functions.
- Define combining forms for organs and know the meaning of related terminology.
- Describe signs, symptoms, and disease conditions affecting the digestive system.

Chapter 5

Lesson 5.1

Introduction

The digestive or gastrointestinal system performs four main functions:

- ingestion
- digestion
- absorption
- elimination

Introduction:

Ingestion and Digestion

- **Ingestion**—food material taken into mouth
- **Digestion**—food is broken down, mechanically and chemically, as it travels through the gastrointestinal tract.
 - Digestive enzymes aid the breakdown of complex nutrients.
 - Proteins → amino acids
 - Sugars → glucose
 - Fats → fatty acids or triglycerides

Introduction:

Absorption

- Digested food passes into the bloodstream through lining cells of the small intestine.
- Nutrients travel to all cells of the body.
- Cells burn nutrients to release the energy stored in food.

Introduction: Elimination

- The body eliminates solid waste materials that cannot be absorbed into bloodstream.
- The large intestine concentrates feces.
- The wastes pass out of the body through the anus.

Organs of the Digestive System

- The gastrointestinal tract begins with the oral cavity.

Oral Cavity (slide 1 of 4)

Major parts of the oral cavity:

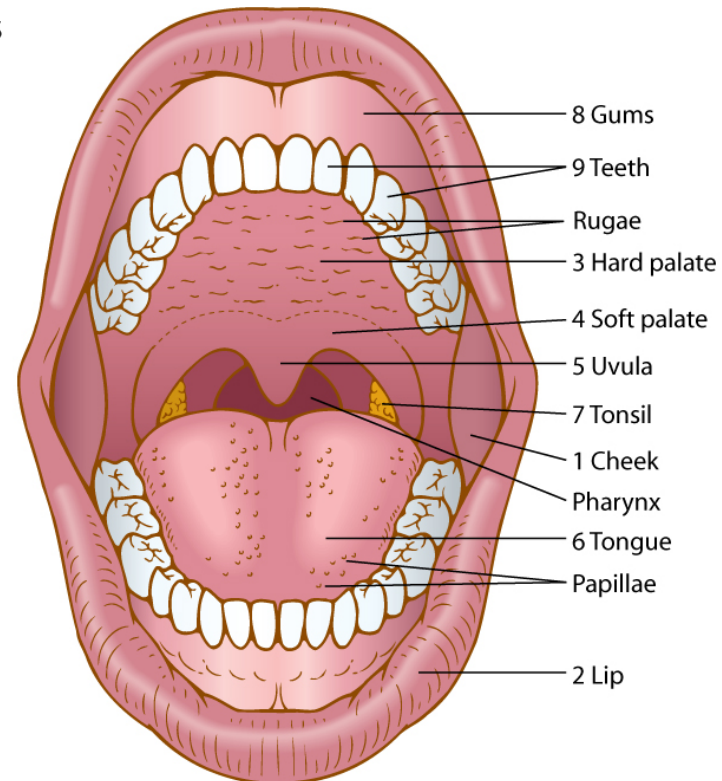
- Cheeks
- Lips
- Hard palate
- Soft palate
- Rugae
- Uvula
- Tongue
- Papillae
- Tonsils
- Gums
- Teeth
- Pharynx

Oral Cavity (slide 2 of 4)

- **Major parts of the oral cavity**

COMBINING FORMS

1. bucc/o
2. cheil/o, labi/o
3. palat/o
4. palat/o
5. uvul/o
6. gloss/o, lingu/o
7. tonsill/o
8. gingiv/o
9. dent/i, odont/o



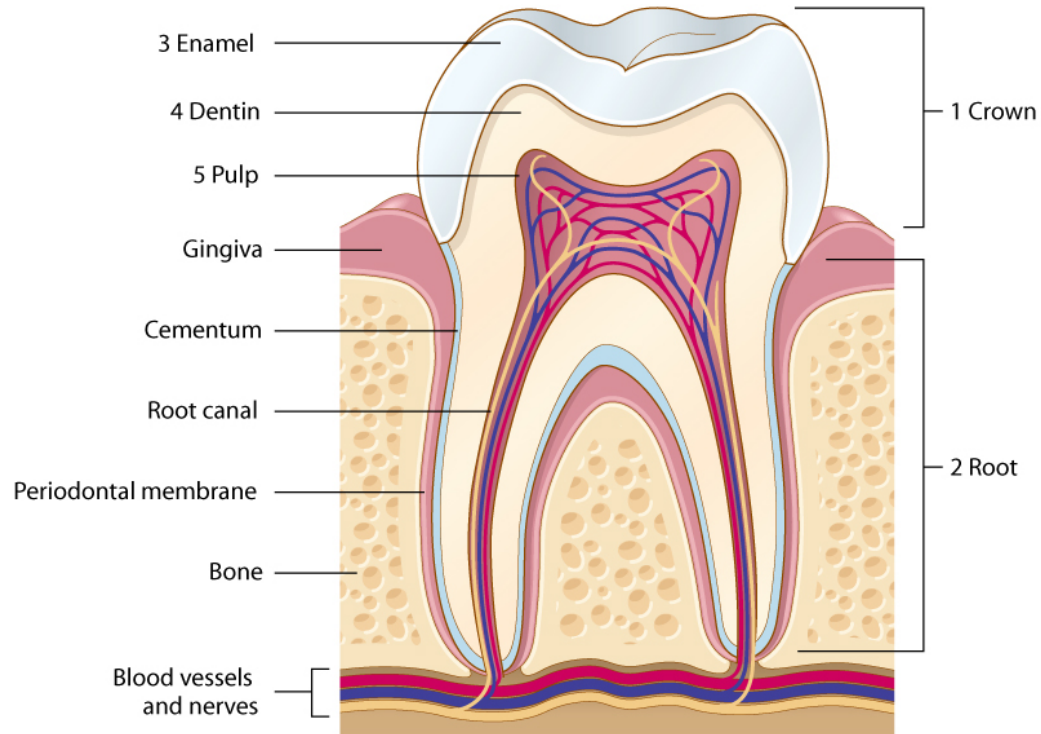
Oral Cavity (slide 3 of 8)

Anatomical terms of a tooth:

- **Crown**
- **Root**
- **Enamel**
- **Dentin**
- **Pulp**
- **Gingiva**
- **Cementum**
- **Root canal**
- **Periodontal membrane**

Oral Cavity (slide 4 of 4)

- **Anatomy of a tooth**



Deglutition (Swallowing)

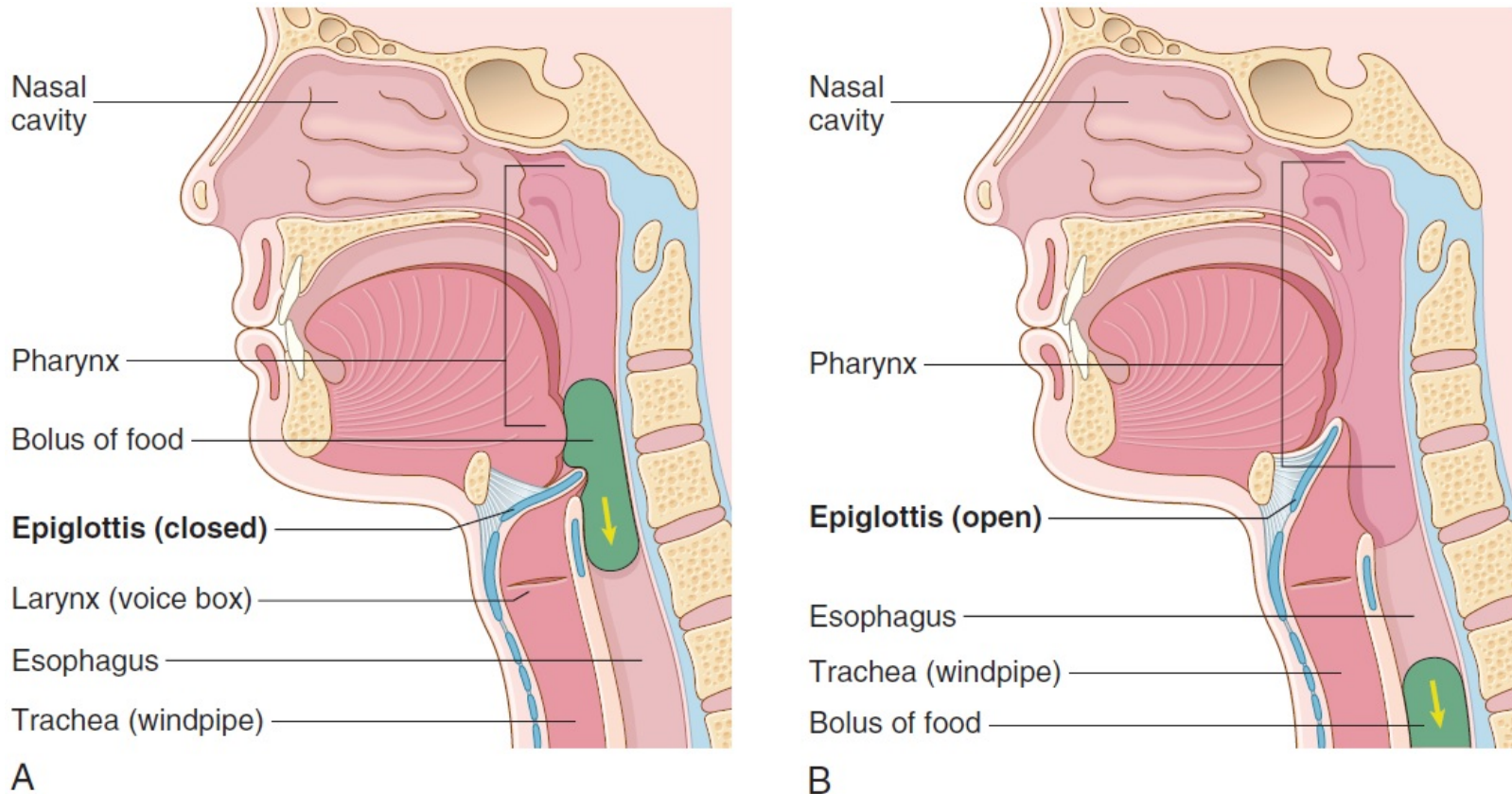


FIGURE 5-5 Deglutition (swallowing). **A**, Epiglottis closes over the trachea as the bolus of food passes down the pharynx toward the esophagus. **B**, Epiglottis opens as the bolus moves down the esophagus.

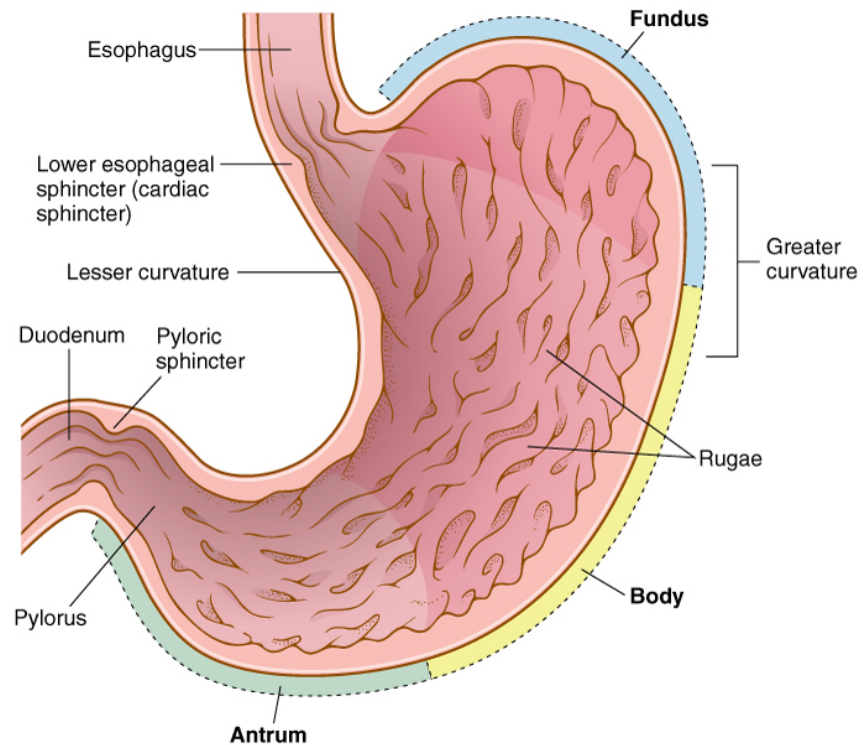
Esophagus/Stomach (slide 1 of 2)

Parts of the stomach include:

- **Lower esophageal sphincter (LES)**
- **Fundus**
- **Greater curvature**
- **Lesser curvature**
- **Rugae**
- **Body**
- **Antrum**
- **Pylorus**
- **Pyloric sphincter**

Esophagus/Stomach (slide 2 of 2)

Parts of the stomach

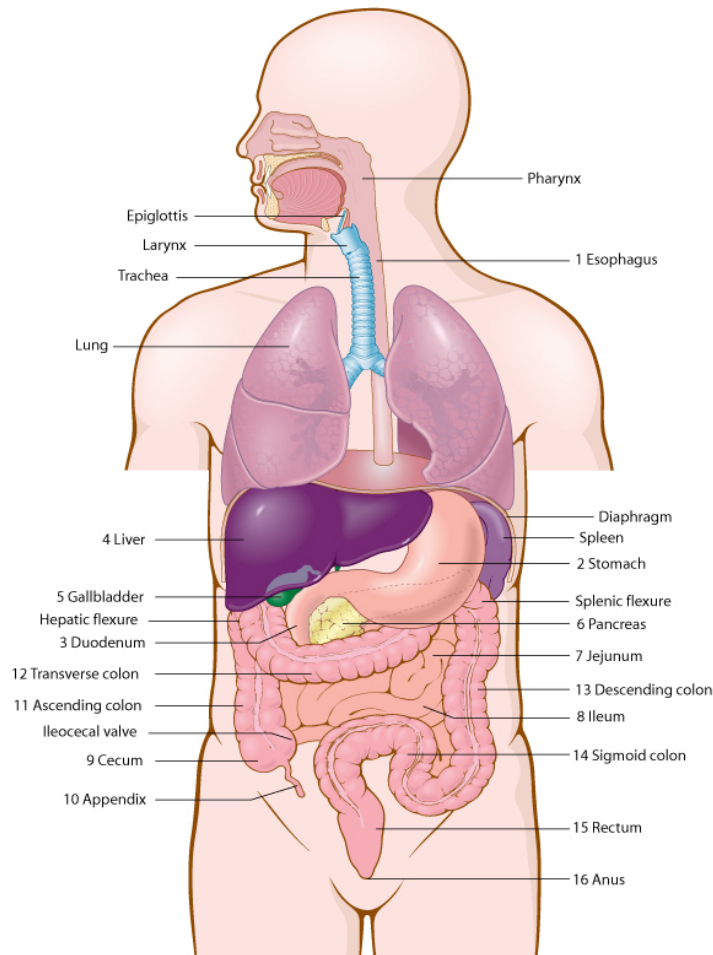


The Digestive Tract (slide 1 of 2)

Anatomical terms of the digestive tract:

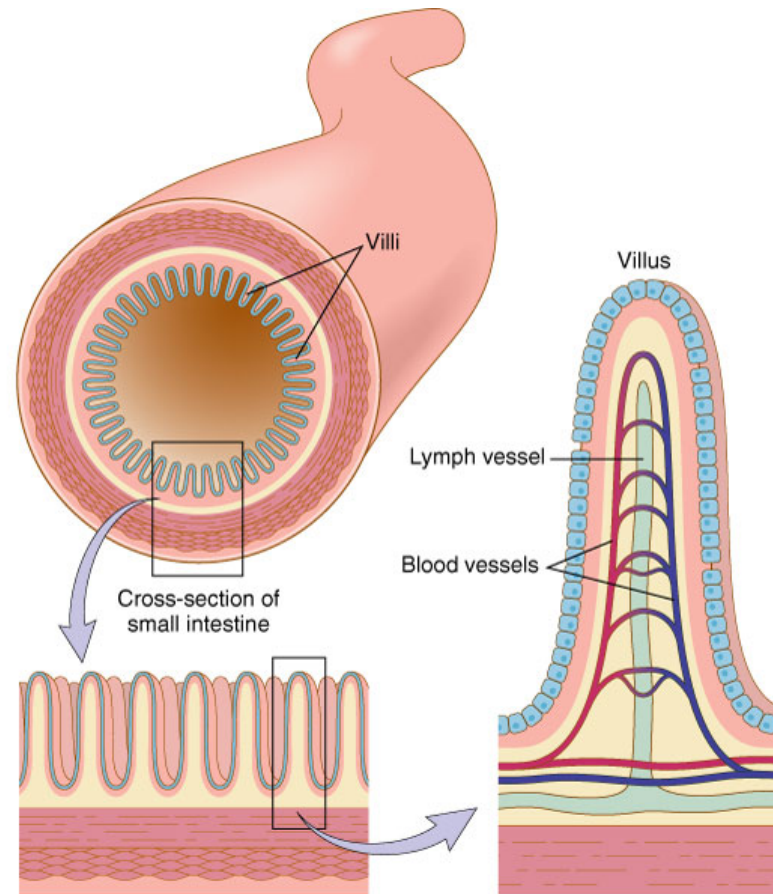
- ☐ Esophagus
- ☐ Stomach
- ☐ Duodenum
- ☐ Liver
- ☐ Gallbladder
- ☐ Pancreas
- ☐ Jejunum
- ☐ Ileum
- ☐ Cecum
- ☐ Appendix
- ☐ Ascending colon
- ☐ Transverse colon
- ☐ Descending colon
- ☐ Sigmoid colon
- ☐ Rectum
- ☐ Anus

The Digestive Tract (slide 2 of 2)



Small Intestine

Villi in the lining of the small intestine



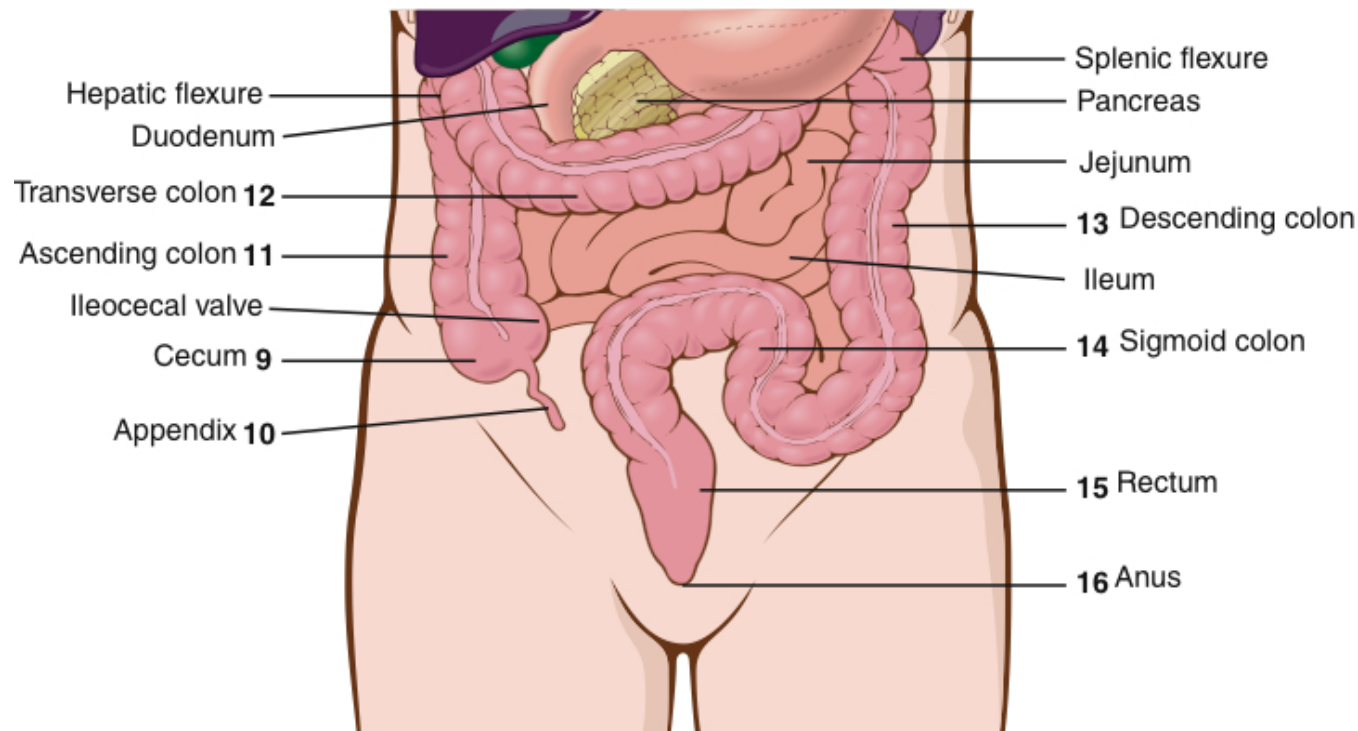
Large Intestine (slide 1 of 2)

Parts of the large intestine:

- Cecum
- Appendix
- Ascending colon
- Transverse colon
- Descending colon
- Sigmoid colon
- Rectum
- Anus

Large Intestine (slide 2 of 2)

Parts of the large intestine



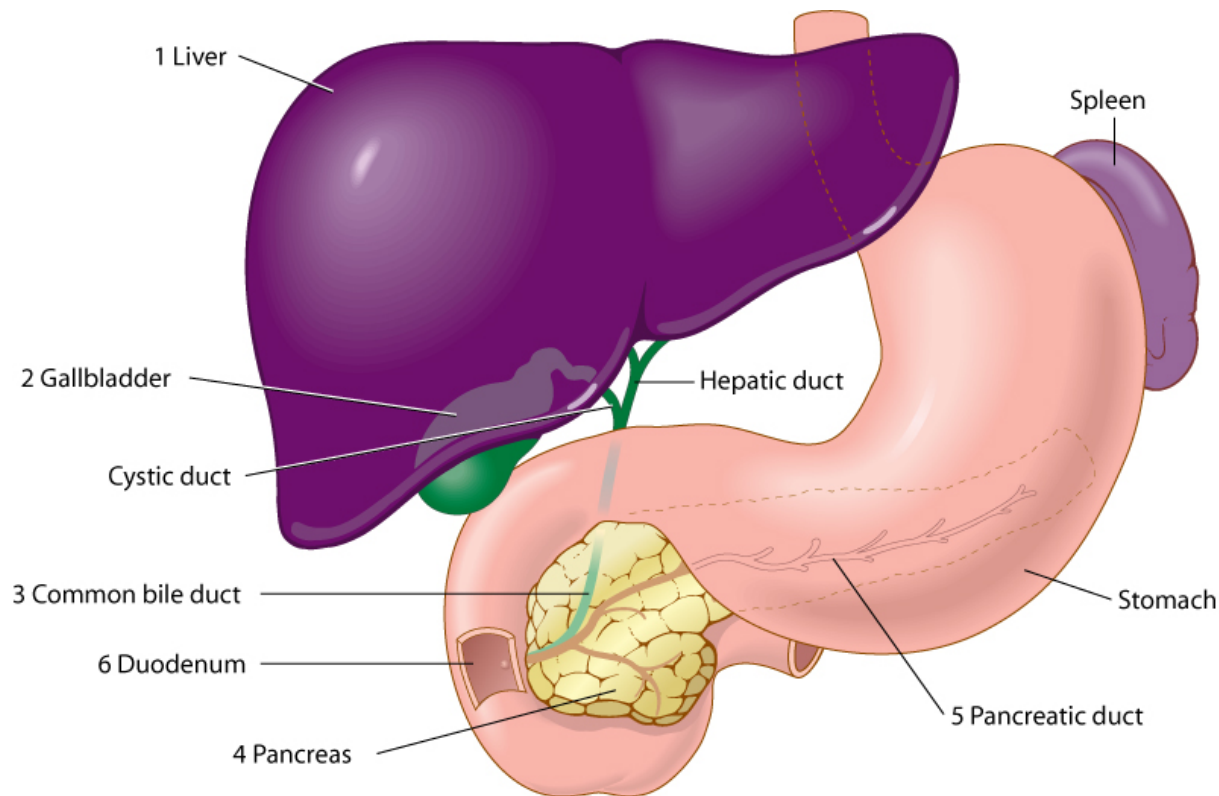
Liver, Gallbladder, and Pancreas (Slide 1 of 5)

Parts of the liver, gallbladder, and pancreas:

- **Liver**
- **Gallbladder**
- **Common bile duct**
- **Pancreas**
- **Pancreatic duct**
- **Duodenum**

Liver, Gallbladder, and Pancreas (slide 2 of 5)

Parts of the liver, gallbladder, and pancreas



Liver, Gallbladder, and Pancreas (slide 3 of 5)

Besides producing bile, the liver:

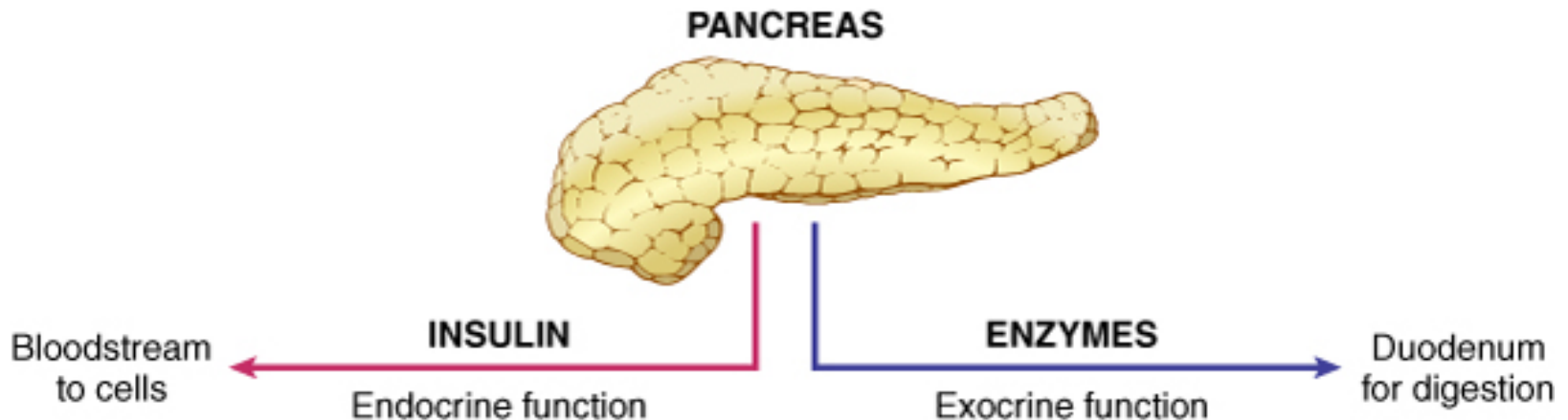
- Helps maintain normal blood glucose levels
- Manufactures blood proteins necessary for clotting
- Releases bilirubin, a pigment in bile
- Removes toxins and poisons from the blood

QUICK QUIZ (Slide 1 of 4)

- 1. What happens if bilirubin cannot leave the body and remains in the bloodstream?**
 - A. Gluconeogenesis
 - B. Emulsification
 - C. Hyperbilirubinemia
 - D. Glycogenolysis

Liver, Gallbladder, and Pancreas (slide 5 of 5)

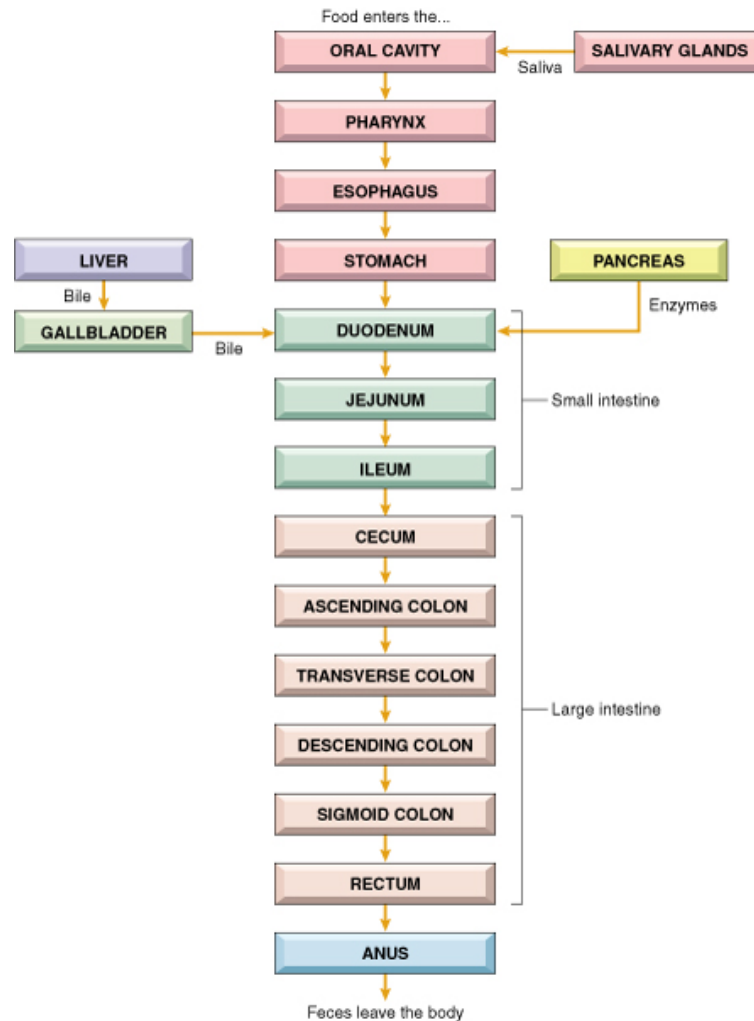
The pancreas and its functions



Food Pathway through the GI Tract (slide 1 of 2)

**Food enters through the oral cavity and
exits through the anus**

Food Pathway through the GI Tract (slide 2 of 2)



QUICK QUIZ (Slide 2 of 4)

2. Which term is the first part of the large intestine?

- A. Cecum
- B. Duodenum
- C. Jejunum
- D. Pylorus

QUICK QUIZ (Slide 3 of 4)

3. Which term means swallowing?

- A. Mastication
- B. Deglutition
- C. Emulsification
- D. Peristalsis

Vocabulary (Slide 1 of 15)

Term	Meaning/Definition
absorption	Passage of materials through the walls of the small intestine into the bloodstream
amino acids	Small building blocks of proteins; released when proteins are digested
amylase	Enzyme secreted by the pancreas and salivary glands to digest starch
anus	Terminal end or opening of the digestive tract to the outside of body

Vocabulary (Slide 2 of 15)

Term	Meaning/Definition
appendix	Blind pouch hanging from the cecum
bile	Digestive juice made in the liver and stored in the gallbladder; breaks up large fat globules; composed of bile pigments, cholesterol, and bile salts
bilirubin	Pigment released by the liver in bile
bowel	Intestine

Vocabulary (Slide 3 of 15)

Term	Meaning/Definition
canine teeth	Pointed, dog-like teeth next to the incisors; also called cuspids or eyeteeth
cecum	First part of the large intestine
colon	Consists of the ascending, transverse, descending, and sigmoid segments
common bile duct	Carries bile from the liver and gallbladder to the duodenum

Vocabulary (Slide 4 of 15)

Term	Meaning/Definition
defecation	Elimination of feces from the digestive tract through the anus
deglutition	Swallowing
dentin	Primary material found in teeth; covered by the enamel in the crown and a protective layer of cementum in the root
digestion	Breakdown of complex foods to simpler forms

Vocabulary (Slide 5 of 15)

Term	Meaning/Definition
duodenum	First part of the small intestine; measures 12 inches long
elimination	Act of removal of materials from the body
emulsification	Physical process of breaking up large fat globules into smaller globules
enamel	Hard, outermost layer of a tooth

Vocabulary (Slide 6 of 15)

Term	Meaning/Definition
enzyme	Chemical that speeds up reactions between substances; enzyme names end in –ase
esophagus	Tube connecting the throat to the stomach
fatty acids	Substances produced when fats are digested; a category of lipids
feces	Solid wastes; stool

Vocabulary (Slide 7 of 15)

Term	Meaning/Definition
gallbladder	Small sac under the liver; stores bile
glucose	Simple sugar
glycogen	Starch; glucose is stored in the form of glycogen in liver cells
hydrochloric acid	Substance produced in the stomach; necessary for digestion of food

Vocabulary (Slide 8 of 15)

Term	Meaning/Definition
ileum	Third part of the small intestine
incisor	One of four front teeth in the dental arch
insulin	Hormone produced by endocrine cells of the pancreas; transports sugar from the blood into cells and stimulates glycogen formation by the liver
jejunum	Second part of the small intestine

Vocabulary (Slide 9 of 15)

Term	Meaning/Definition
lipase	Pancreatic enzyme needed to digest fats
liver	Large organ located in the RUQ of the abdomen; secretes bile; stores sugar, iron, and vitamins; produces blood proteins; destroys worn-out RBCs; filters out toxins; normal adult liver weighs about 2 ½ to 3 pounds
lower esophageal sphincter (LES)	Ring of muscles between the esophagus and stomach

Vocabulary (Slide 10 of 15)

Term	Meaning/Definition
mastication	Chewing
molar teeth	Sixth, seventh, and eighth teeth from the middle on either side of the dental arch
palate	Roof of the mouth
pancreas	Organ behind the stomach; produces insulin and enzymes

Vocabulary (Slide 11 of 15)

Term	Meaning/Definition
papillae (<i>singular: papilla</i>)	Small, projections on the tongue
parotid gland	Salivary gland within the cheek, just anterior to the ear
peristalsis	Rhythmic contractions of the tubular organs
pharynx	Throat, the common passageway for food from the mouth and for air from the nose

Vocabulary (Slide 12 of 15)

Term	Meaning/Definition
portal vein	Large vein bringing blood to the liver from the intestines
protease	Enzyme that digests protein
pulp	Soft tissue within a tooth, containing nerves and blood vessels
pyloric sphincter	Ring of muscle at the end of the stomach, near the duodenum; opens when a wave of peristalsis passes over it

Vocabulary (Slide 13 of 15)

Term	Meaning/Definition
pylorus	Distal region of the stomach, opening to the duodenum
rectum	Last section of the large intestine, connecting the end of the colon and the anus
rugae	Ridges on the hard palate and the wall of the stomach
saliva	Digestive juice produced by salivary glands; contains the enzyme amylase

Vocabulary (Slide 14 of 15)

Term	Meaning/Definition
salivary glands	Parotid, sublingual, and submandibular glands
sigmoid colon	Fourth and last, S-shaped segment of the colon, just before the rectum; empties into the rectum
sphincter	Circular ring of muscle that constricts a passage or closes a natural opening
stomach	Muscular organ that receives food from the esophagus

Vocabulary (Slide 15 of 15)

Term	Meaning/Definition
triglycerides	Fat molecules composed of three parts fatty acids and one part glycerol; subgroup of lipids
uvula	Soft tissue hanging from the middle of the soft palate
villi (<i>singular: villus</i>)	Microscopic projections in the wall of the small intestine that absorb nutrients into the bloodstream

Terminology – Combining Forms

Parts of the Body (Slide 1 of 5)

Combining Form	Meaning
an/o	anus
append/o, appendic/o	appendix
bucc/o	cheek
cec/o	cecum
celi/o	belly, abdomen
cheil/o	lip
cholecyst/o	gallbladder
choledoch/o	common bile duct

Terminology – Combining Forms

Parts of the Body (Slide 2 of 5)

Combining Form	Meaning
col/o	colon
colon/o	colon
dent/i	tooth
duoden/o	duodenum
enter/o	intestines, usually small intestine
esophag/o	esophagus
faci/o	face
gastr/o	stomach
gingiv/o	gums

Terminology – Combining Forms

Parts of the Body (Slide 3 of 5)

Combining Form	Meaning
gloss/o	tongue
hepat/o	liver
ile/o	ileum
jejun/o	jejunum
labi/o	lip
lapar/o	abdomen
lingu/o	tongue
mandibul/o	lower jaw, mandible
odont/o	tooth

Terminology – Combining Forms

Parts of the Body (Slide 4 of 5)

Combining Form	Meaning
or/o	mouth
palat/o	palate
pancreat/o	pancreas
peritone/o	peritoneum
pharyng/o	throat
proct/o	anus and rectum
pylor/o	pyloric sphincter
rect/o	rectum
sialaden/o	salivary gland

Terminology – Combining Forms

Parts of the Body (Slide 5 of 5)

Combining Form	Meaning
sigmoid/o	sigmoid colon
stomat/o	mouth
uvul/o	uvula

Terminology – Combining Forms

Substances (Slide 1 of 2)

Combining Form	Meaning
amyl/o	starch
bil/i	gall, bile
bilirubin/o	bilirubin (bile pigment)
chol/e	gall or bile
chlorhydr/o	hydrochloric acid
gluc/o	sugar
glyc/o	sugar

Terminology – Combining Forms

Substances (Slide 2 of 2)

Combining Form	Meaning
glycogen/o	glycogen, animal starch
lip/o	fat
lith/o	stone
prote/o	protein
py/o	pus
sial/o	saliva, salivary
steat/o	fat

Terminology – Suffixes

Suffix	Meaning
-ase	enzyme
-chezia	defecation, elimination of wastes
-iasis	abnormal condition
-prandial	meal

Anastomoses

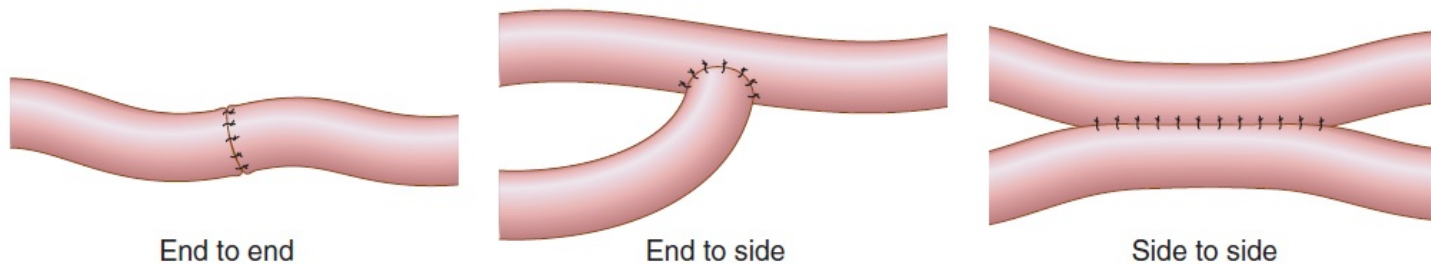


FIGURE 5-16 Three types of anastomoses. These are examples of an enteroenterostomy. The suffix -stomy, when used with two or more combining forms (enter/o and enter/o), indicates the surgical creation of a new opening between those parts of the body.

QUICK QUIZ (Slide 4 of 4)

4. Which term means inflammation of the lip?

- A. Cholecystitis
- B. Celiac
- C. Appendicitis
- D. Cheilitis

Chapter 5

Lesson 5.2

Pathology

of the Digestive System (slide 1 of 3)

Signs and symptoms

- **Anorexia** – lack of appetite
- **Ascites** – abnormal accumulation of fluid in the abdomen
- **Borborygmi** (*singular: borborygmus*) – rumbling or gurgling noises produced by the movement of gas, fluid, or both in the GI tract
- **Constipation** – difficulty in passing stools
- **Diarrhea** – frequent passage of loose, watery stools

Pathology

of the Digestive System (slide 2 of 3)

- **Dysphagia** – difficulty in swallowing
- **Eructation** – gas expelled from the stomach through the mouth
- **Flatus** – gas expelled through the anus
- **Hematochezia** – passage of fresh, bright red blood from the rectum
- **Jaundice (icterus)** – yellow-orange coloration of the skin and whites of the eyes caused by high levels of bilirubin in the blood

Pathology

of the Digestive System (slide 3 of 3)

- **Melena** – black, tarry stools; feces containing digested blood
- **Nausea** – unpleasant sensation in the stomach with a tendency to vomit
- **Steatorrhea** – fat in the feces; frothy, foul-smelling fecal matter

Pathologic Conditions

Oral Cavity and Teeth

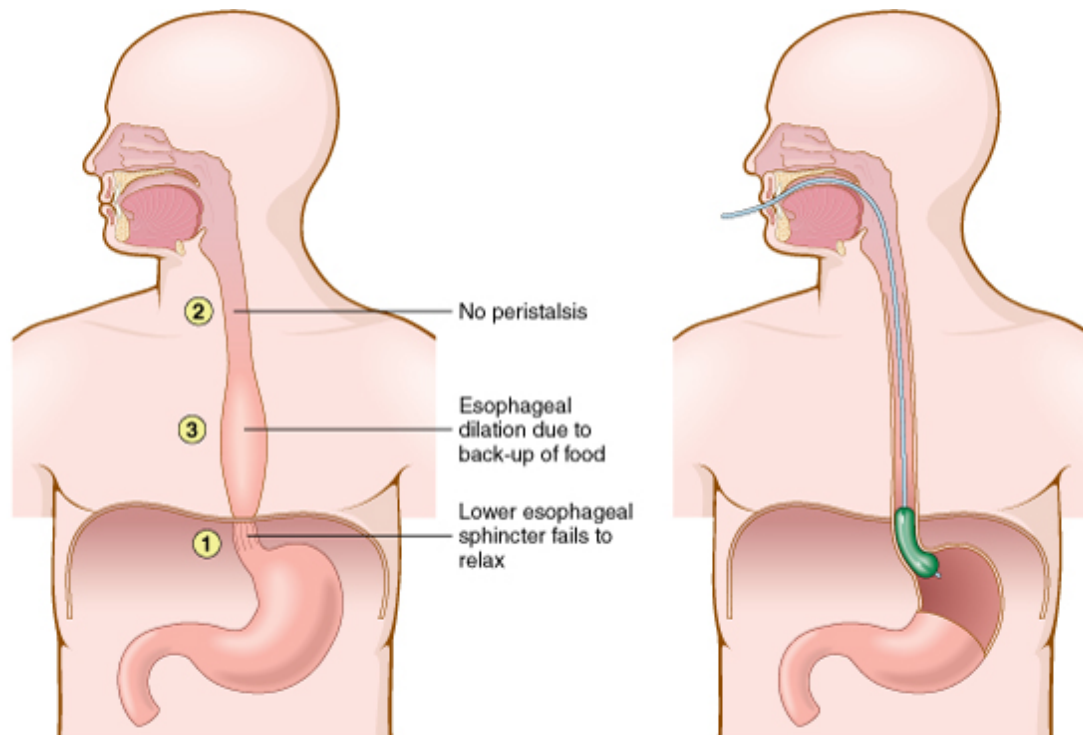
- **Aphthous stomatitis** – inflammation of the mouth with small, painful ulcers
- **Dental caries** – tooth decay
- **Herpetic stomatitis** – inflammation of the mouth by infection with the herpes virus
- **Oral leukoplakia** – white plaques or patches
- **Periodontal disease** – inflammation and degeneration of the gums, teeth and surrounding bone

Pathologic Conditions

Upper Gastrointestinal Tract

(slide 1 of 5)

- **Achalasia** – failure of the lower esophagus sphincter (LES) muscle to relax



Pathologic Conditions

Upper Gastrointestinal Tract

(slide 2 of 5)

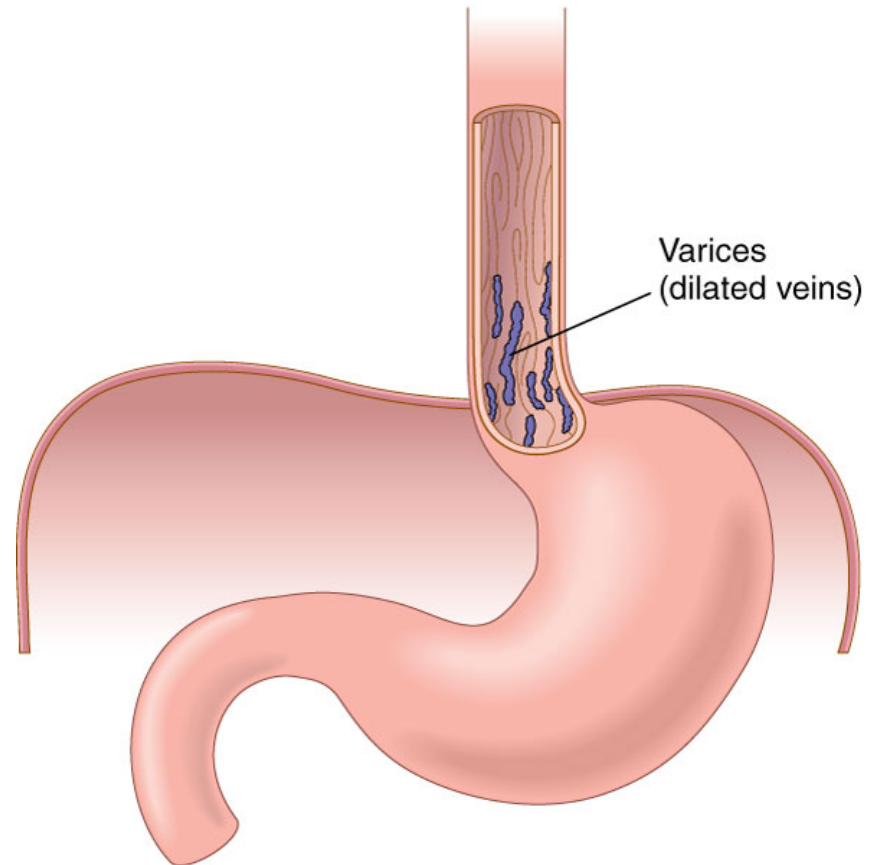
- **Esophageal cancer** – malignant tumor of the esophagus
- **Gastric carcinoma** – malignant tumor of the stomach

Pathologic Conditions

Upper Gastrointestinal Tract

(slide 3 of 5)

- **Esophageal varices** – swollen, varicose veins at the lower end of the esophagus



Pathologic Conditions

Upper Gastrointestinal Tract

(slide 4 of 5)

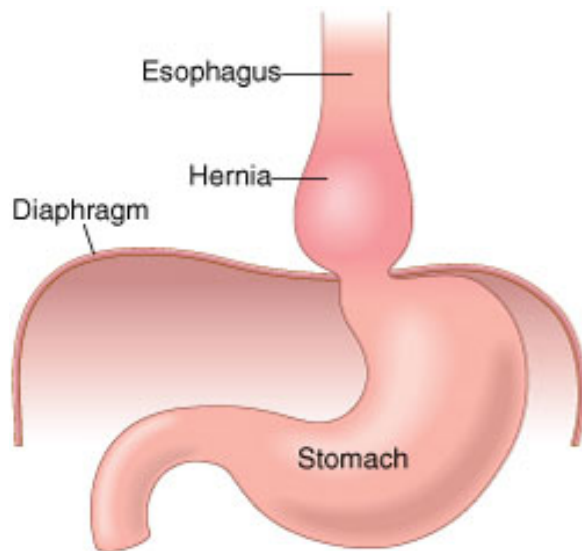
- **Gastroesophageal reflux disease (GERD)** – solids and fluids return to the mouth from the stomach
- **Peptic ulcer** – open sore or lesion of the mucous membrane of the stomach or duodenum

Pathologic Conditions

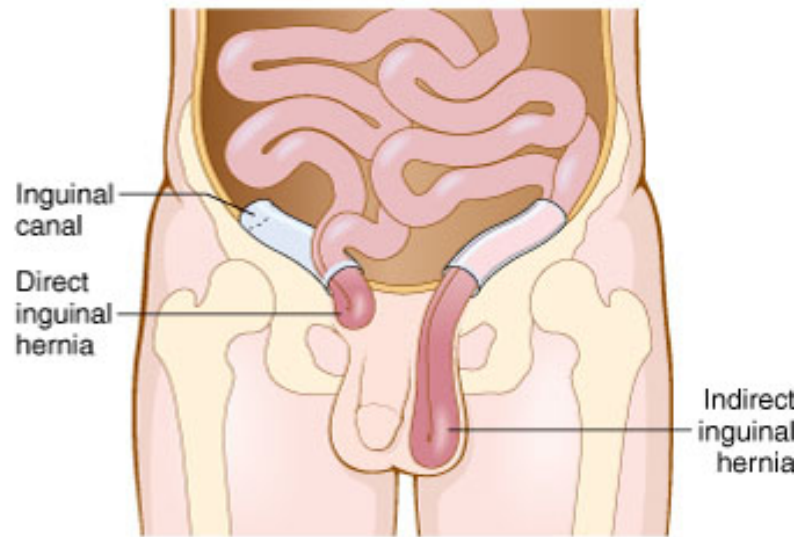
Upper Gastrointestinal Tract

(slide 5 of 5)

- **Hernia** – protrusion of an organ or part through the muscle normally containing it



HIATAL (DIAPHRAGMATIC) HERNIA



INGUINAL HERNIAS

Pathologic Conditions

Lower Gastrointestinal Tract

Small and Large Intestines (slide 1 of 5)

- **Anal fistula** – abnormal tubelike passageway near the anus
- **Colonic polyposis** – polyps protrude from the mucous membrane of the colon

Pathologic Conditions

Lower Gastrointestinal Tract

Small and Large Intestines (slide 2 of 5)

- **Colorectal cancer** – adenocarcinoma of the colon or rectum or both

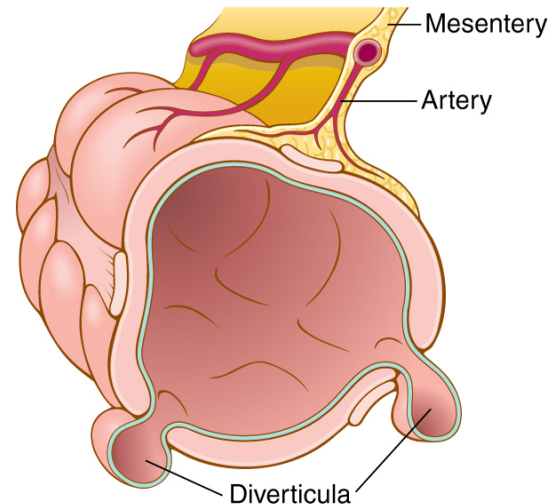


Pathologic Conditions

Lower Gastrointestinal Tract

Small and Large Intestines (slide 3 of 5)

- **Crohn disease** – chronic inflammation of the intestinal tract
- **Diverticulosis** – abnormal side pockets (outpouchings) in the intestinal wall



Pathologic Conditions

Lower Gastrointestinal Tract

Small and Large Intestines (slide 4 of 5)

- **Dysentery** – painful, inflamed intestines
- **Hemorrhoids** – swollen, twisted, varicose veins in the rectal region
- **Ileus** – failure of peristalsis with resulting obstruction of the intestines
- **Inflammatory bowel disease (IBD)** – inflammation of the colon and small intestine

Pathologic Conditions

Lower Gastrointestinal Tract

Small and Large Intestines (slide 5 of 5)

- **Intussusception** – telescoping of the intestines
- **IBS** – irritable bowel syndrome – group of gastrointestinal symptoms associated with stress and tension
- **Ulcerative colitis** – chronic inflammation of the colon with the presence of ulcers
- **Volvulus** – twisting of the intestines on itself

Pathologic Conditions

Liver Gallbladder, and Pancreas (slide 1 of 2)

- **Cirrhosis** – chronic degenerative disease of the liver
- **Hepatocellular carcinoma (HCC)** – liver cancer
- **Pancreatic cancer** – malignant tumor of the pancreas
- **Pancreatitis** – inflammation of the pancreas
- **Viral hepatitis** – inflammation of the liver caused by a virus

Pathologic Conditions

Liver Gallbladder, and Pancreas (slide 2 of 2)

- **Cholelithiasis** – gallstones in the gallbladder

